

# A cofinal bound for maximum-modulus points of entire functions

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## Abstract

For every nonzero entire function which is not a monomial, we prove that the number of its exact maximum-modulus points is bounded by a fixed finite integer at arbitrarily large radii. This gives a negative answer to the all-radii maximum-modulus question. After removing the zero at the origin, one may take the bound to be twice the order at zero of the normalized logarithmic derivative minus its constant term. The proof uses finite local inverse-root families and continuation of the closure of regular targets with large fibers. It handles the omitted logarithmic-derivative value and the value at infinity without assuming global normalization or a generic component degree. A complete Lean 4 formalization accompanies the proof.

## 1 The problem and the result

Throughout,  $\mathbb{N} = \{0, 1, 2, \dots\}$ , so the monomials excluded below include nonzero constant functions.

For an entire function  $f$  and a positive radius  $r$ , write

$$M_f(r) = \max_{|z|=r} |f(z)|.$$

$$\text{MaxPoints}(f, r) = \{z \in \mathbb{C} : |z| = r \text{ and } |f(w)| \leq |f(z)| \text{ for every } w \text{ with } |w| = r\}.$$

Thus the maximum-point set is defined by actual function values. The maximum exists by compactness. For a nonzero non-monomial entire function, we prove below that this set is finite at every positive radius; its cardinality  $\nu_f(r)$  consequently has its ordinary finite meaning.

Hayman–Lingham’s Problem 2.16(b), on printed page 29 of [1], asks whether a single entire function can have  $\liminf_{r \rightarrow \infty} \nu_f(r) = \infty$ . The non-monomial formulation is explicit in Glücksam–Pardo–Simón’s Question 1.1(b) [3] and in the introduction of Pardo–Simón–Sixsmith [4]. In contrast, Herzog–Piranian [2] answered the selected-radii question  $\limsup_{r \rightarrow \infty} \nu_f(r) = \infty$  affirmatively in 1968. These two questions have different quantifiers. The recent finite-order result [4] concerns the latter question. Approximate maximum points, as studied in [3], also differ from the exact set used here.

**Theorem 1.** *Let  $f$  be a nonzero entire function which is not  $cz^m$  for any  $c \neq 0$  and  $m \in \mathbb{N}$ . There is an integer  $B \in \mathbb{N}$  such that, for every  $R > 0$ , there is an  $r > R$  for which  $\text{MaxPoints}(f, r)$  is finite and  $\text{card } \text{MaxPoints}(f, r) \leq B$ . In fact, write  $f(z) = z^m g(z)$  with  $g(0) \neq 0$ . One may take*

$$B = 2k, \quad k = \text{ord}_0 \left( \frac{zg'(z)}{g(z)} \right).$$

*Here  $k$  is a positive finite integer.*

**Corollary 2.** *No nonzero non-monomial entire function has  $\nu_f(r) \rightarrow \infty$  as  $r \rightarrow \infty$  along all positive radii.*

*Deduction from Theorem 1.* Fix the bound  $B$  supplied by the theorem. If the count tended to infinity, there would be a threshold  $R_0$  beyond which it was at least  $B + 1$ . Choose  $R > \max\{0, R_0\}$  and then the radius  $r > R$  supplied by the theorem. Its maximum-point set is finite and has cardinality at most  $B$ , a contradiction. The same argument rules out a formulation which treats an infinite maximum-point set as having infinite cardinality.  $\square$

The proof first bounds fibers with a small coordinate. It then constructs finite local covers of the actual logarithmic-derivative correspondence, counting physical pairs rather than inverse-root indices. The closure of the regular targets with too many pairs continues across every spherical value, including an omitted value. Its product projection would be both open and closed in the plane, contradicting the small-coordinate bound. A countable exceptional set and the radial maximum envelope then give a good radius in every positive interval.

## 2 Factorization and finiteness on each circle

Factor  $f(z) = z^m g(z)$ , where  $m \in \mathbb{N}$ ,  $g$  is entire, and  $g(0) \neq 0$ . The factor  $g$  is nonconstant, since otherwise  $f$  is a monomial. Put

$$g^\sharp(z) = \overline{g(\bar{z})}.$$

This reflection is entire. On every positive circle,  $|f(z)| = r^m |g(z)|$ , so  $f$  and  $g$  have the same maximum-point set.

**Lemma 3.** *For each  $s \in \mathbb{C}^\times$ , the function  $P_s(z) = g(z)g^\sharp(s/z)$  is holomorphic and nonconstant on  $\mathbb{C}^\times$ .*

*Proof.* Holomorphy follows by composition and multiplication. If  $P_s$  were a constant  $C$ , then, as  $z \rightarrow \infty$ , the factor  $g^\sharp(s/z)$  would tend to  $g^\sharp(0) \neq 0$ . Outside a sufficiently large disk it would be bounded away from zero, and the identity  $g(z) = C/g^\sharp(s/z)$  would bound  $g$ . Continuity bounds  $g$  on the remaining compact disk. Liouville's theorem would make  $g$  constant, contrary to its choice.  $\square$

**Lemma 4.** *The set  $\text{MaxPoints}(f, r)$  is finite for every  $r > 0$ .*

*Proof.* It suffices to consider  $g$ . If its maximum-point set on the compact circle were infinite, it would have an accumulation point on that circle, hence in  $\mathbb{C}^\times$ . For every maximum point  $z$ , the identity  $r^2/z = \bar{z}$  gives

$$P_{r^2}(z) = g(z)g^\sharp(\bar{z}) = |g(z)|^2 = M_g(r)^2.$$

The identity theorem on the connected domain  $\mathbb{C}^\times$  would make  $P_{r^2}$  constant. This contradicts Lemma 3.  $\square$

This proves finiteness independently of the subsequent fiber bound and justifies using  $\nu_f(r) = \text{card MaxPoints}(f, r)$  throughout the positive domain.

### 3 Logarithmic derivatives and the small-coordinate bound

Define

$$A(z) = m + \frac{zg'(z)}{g(z)}, \quad B(z) = A^\sharp(z).$$

These are meromorphic functions interpreted as continuous functions into  $\widehat{\mathbb{C}}$ . The point zero is regular and  $A(0) = B(0) = m$ . At a zero  $\alpha \neq 0$  of  $g$  of multiplicity  $n$ , write  $g(z) = (z - \alpha)^n h(z)$  with  $h(\alpha) \neq 0$ . Then

$$A(z) = \frac{n\alpha}{z - \alpha} + m + n + \frac{zh'(z)}{h(z)}.$$

Thus every pole is genuine and simple, and its spherical value is  $\infty$ . The same holds for  $B$ .

The function  $A - m$  is analytic near zero and not identically zero. Indeed, if it vanished on a neighborhood,  $g'$  would vanish on a punctured neighborhood where  $g \neq 0$ , and the identity theorem would make  $g$  constant. Let

$$k = \text{ord}_0(A - m) > 0, \quad L = 2k.$$

Reflection gives the same positive finite order for  $B - m$ .

For any product  $s$  and spherical common value  $p$ , let

$$F(s, p) = \{(z, w) \in \mathbb{C}^2 : zw = s, A(z) = B(w) = p\}.$$

We count distinct actual ordered pairs.

We will repeatedly use the following local power coordinate. If a nonconstant analytic function  $h$  satisfies  $h(\alpha) = p_0$ , write

$$h(z) - p_0 = (z - \alpha)^d u(z), \quad u(\alpha) \neq 0.$$

After shrinking to a simply connected neighborhood,  $u$  has an analytic  $d$ th root. The coordinate  $e(z) = (z - \alpha)u(z)^{1/d}$  has  $e'(\alpha) \neq 0$ , so the inverse function theorem makes it conformal and  $h = p_0 + e^d$ . In that coordinate, each value has at most  $d$  distinct inverse roots. Apply this to  $A - m$  and  $B - m$  at zero. Choose  $\varepsilon > 0$  such that both coordinate disks have this  $k$ -root bound and contain no poles.

For  $s \neq 0$ , a pair in  $F(s, p)$  is determined by either coordinate, since  $w = s/z$  and  $z = s/w$ . Thus at most  $k$  pairs have  $|z| < \varepsilon$ , and at most  $k$  pairs have  $|w| < \varepsilon$ . Their union has at most  $L$  pairs, also when  $p = \infty$ . In particular,

$$0 < |s| < \varepsilon^2 \implies F(s, p) \text{ is finite and } \text{card } F(s, p) \leq L \text{ for every } p \in \widehat{\mathbb{C}}. \quad (3.1)$$

Indeed, a pair whose two coordinate norms were at least  $\varepsilon$  would have  $|zw| \geq \varepsilon^2$ .

### 4 Properness and exact finite local covers

Consider the actual correspondence

$$X = \{(z, w) \in \mathbb{C}^\times \times \mathbb{C}^\times : A(z) = B(w) \neq m\}$$

over

$$\Omega = \mathbb{C}^\times \times (\widehat{\mathbb{C}} \setminus \{m\}),$$

and the map  $\Phi : X \rightarrow \Omega$ ,  $\Phi(z, w) = (zw, A(z))$ . It is proper with finite fibers. Indeed, on a compact target set the common values stay away from  $m$ . Continuity of  $A, B$  at zero gives  $\eta > 0$  bounding both coordinate norms below. A bound  $|zw| \leq S$  then bounds both norms

above by  $S/\eta$ . The inverse image is a closed subset of the resulting compact annulus product. For a fixed finite value the inverse roots are discrete, by nonconstancy and the identity theorem; at infinity they are isolated poles. Hence there are finitely many in the compact annulus, and the product fixes the other coordinate. Denote the actual target image in  $\Omega$  by  $Y$ . Properness makes  $Y$  closed in  $\Omega$ .

All local target neighborhoods in this section lie inside  $\Omega$ . Fix  $y_0 = (s_0, p_0) \in \Omega$ . Its actual central fiber is finite. If it is empty, properness gives a target neighborhood disjoint from  $Y$ , so the local cover there is empty. Otherwise, properness captures every nearby pair in a finite collection of chart products around the central pairs. To see this, an uncaptured sequence over targets tending to  $y_0$  would have a subsequence converging to a central pair inside one of those chart products.

In a finite-value chart, the analytic power-coordinate construction gives

$$A(z) = p_0 + e_A(z)^{d_A}, \quad B(w) = p_0 + e_B(w)^{d_B},$$

where  $e_A, e_B$  are conformal coordinates centered at their physical roots. Choose one positive common multiple  $N$  of the finitely many orders. For  $p = p_0 + t^N$ , every local inverse root is of the form

$$z = e_A^{-1}(\zeta t^{N/d_A}), \quad w = e_B^{-1}(\xi t^{N/d_B}), \quad \zeta^{d_A} = \xi^{d_B} = 1.$$

This constructs a finite analytic physical family  $P_i(t) = (z_i(t), w_i(t))$ , with product functions  $S_i(t) = z_i(t)w_i(t)$  and raw image maps

$$\gamma_i(t) = (S_i(t), p_0 + t^N). \quad (4.1)$$

All displayed pairs have the actual common values. Every nearby actual pair is represented by an index at that same  $t$ , because the roots of each power equation are exactly the displayed root-of-unity multiples. At zero every central physical pair is represented. Indices need not be injective.

Localize this capture in the target as follows. Choose a product disk about  $s_0$  and a value disk whose product lies in the capture neighborhood. Choose  $\delta > 0$  small enough that all raw product images on  $|t| < \delta$  lie in that product disk and  $\delta^N$  is smaller than the initial value radius. Use the value radius  $\delta^N$  in the final neighborhood  $W$ . The full raw open-disk images then lie in  $W$ , and every target in  $W$  has all its parameter roots inside that same disk. Consequently

$$Y \cap W = \bigcup_i \gamma_i(\{|t| < \delta\}). \quad (4.2)$$

This is an exact cover of the total actual image.

At  $p_0 = \infty$ , use  $\lambda = 1/p$ . Every pole of  $A$  and  $B$  is simple, so  $1/A$  and  $1/B$  have simple zeros at the corresponding physical roots. They are conformal coordinates, and the physical roots are analytic functions of  $t = \lambda$  itself. Here take  $N = 1$  and  $p(t) = 1/t$  for  $t \neq 0$ , with  $p(0) = \infty$ . Properness and the same target shrinking give the exact cover. Its punctured points have finite common value. In either chart type, if  $v$  is the centered value coordinate, then  $|v(p(t))| = |t|^N$ .

## 5 Stabilized branches and distinct physical counts

For finitely many indices  $i, j$  and  $N$ th roots of unity  $\omega$ , the functions

$$S_i(\omega t) - S_j(t)$$

are analytic at zero. By the isolated-zero theorem, after a common shrinking each is either identically zero as a germ or nonzero for every nonzero parameter. Stabilize the comparisons

of both physical coordinates as well. Repeat the matched target shrinking above after this change of radius. These finite comparisons give the following facts.

1. Two punctured raw image representatives are equal or disjoint. An intersection has parameter roots related by  $t \mapsto \omega t$ ; the comparison is therefore a persistent identity. Rotation preserves the parameter disk.
2. Every noncentral point of the finite raw image union has a neighborhood where the whole union is one analytic graph over the common value coordinate. The map  $t \mapsto t^N$  has nonzero derivative there. Coincident rotated roots give the same graph germ by the persistent comparison. Noncoincident roots and the other disjoint branches are absent after shrinking the neighborhood. Relative closedness, proved below, also excludes limits from those other branches.
3. Each punctured image branch has a constant physical fiber cardinal  $K$ . Exact capture at the same parameter gives, for each index  $j$ ,

$$F(S_j(t), p(t)) = \{P_i(t) : S_i(t) = S_j(t)\}. \quad (5.1)$$

The product filter and the equivalence relation given by physical equality  $P_i(t) = P_{i'}(t)$  are constant on the punctured disk, by the stabilized comparisons. This finite image therefore has constant cardinality. It counts distinct pairs even when several indices represent the same pair.

4. The finite raw image union, and any subunion of its branches, is relatively closed in its open value cylinder. If targets converge inside that cylinder,  $|v(p)| = |t|^N$  bounds their parameters strictly inside the disk. Passing to a fixed index and then to a convergent parameter subsequence supplies the target limit, including at the center.

There is an essential central-cardinality comparison. Suppose the entire actual image at a finite center is locally a single graph  $s = \psi(p)$ . Forward validity forces every product germ to be  $\psi(p_0 + t^N)$ , so all indices belong to one persistent product class. Choose index representatives of distinct central physical pairs. If two representatives coincide at a small nonzero parameter, stabilized physical equality makes their germs equal, and hence their central pairs equal. The distinct chosen central representatives thus remain distinct at that parameter. Therefore

$$\text{card } F(s_0, p_0) \leq K. \quad (5.2)$$

This uses the graph property of the total image and the exact cover.

## 6 The regular high-count closure

Call a point  $(s_0, p_0) \in Y$  *regular* if in some product neighborhood  $U_s \times U_p$  the total image satisfies

$$Y \cap (U_s \times U_p) = \{(\psi(p), p) : p \in U_p\}$$

for an analytic function  $\psi$  with values in  $U_s$ . The value coordinate is spherical when needed. This is equality for the whole local image.

Let  $E$  be the set of regular finite-valued targets in  $Y$  whose actual fiber has more than  $L$  distinct pairs, and let

$$H = \text{cl } E \quad \text{in } \mathbb{C} \times \widehat{\mathbb{C}}.$$

By (3.1),  $E$  misses the full open cylinder  $\{|s| < \varepsilon^2\} \times \widehat{\mathbb{C}}$ : its first coordinate is nonzero by definition, and every nonzero product in that cylinder has at most  $L$  pairs. The same open

cylinder is disjoint from  $H$ . In particular, every point of  $H$  has nonzero product coordinate. Closedness of  $Y$  in  $\Omega$  also gives  $H \subset Y$  locally away from  $p = m$ .

In a stabilized local finite family, select the branches whose punctured physical count  $K$  exceeds  $L$ . On the matched neighborhood  $W$ ,

$$\text{selected punctured union} \subset E \cap W \subset \text{selected full union}.$$

Away from the center this follows from the exact whole-image graph and count. A finite central point of  $E$  belongs to a selected branch by (5.2); it cannot contribute an isolated point. At infinity the center itself is not in  $E$ , but the punctured pole-chart points are finite-valued and satisfy the same selection rule. Relative closedness and density of each punctured branch in its full image give

$$H \cap W = \text{selected full union} \quad (p_0 \neq m). \quad (6.1)$$

Closure is localized inside the open neighborhood: every sequence tending to a point of  $W$  is eventually in  $W$ . Each point of  $H$  away from  $m$  therefore lies on a complete local analytic branch contained in  $H$ , including when its value is infinity.

## 7 Continuation across the omitted value

Fix  $y_0 = (s_0, m) \in H$ ; the preceding exclusion gives  $s_0 \neq 0$ . Choose a target neighborhood with  $|s| \leq S$ . A point of  $E$  there has more than  $L$  pairs, of which at most  $L$  have a small coordinate. At least one surviving actual pair satisfies

$$\varepsilon \leq |z|, |w| \leq S/\varepsilon. \quad (7.1)$$

The upper bounds follow from the product bound and the lower bounds. The surviving pair can change from target to target; no persistent choice of it is needed.

The roots of  $A - m$  and  $B - m$  in this compact annulus are finite. Finite-value power charts at these roots construct a finite common-power family across  $p = m$ . Compactness captures all pairs satisfying (7.1) in the chosen chart neighborhoods when  $p$  is sufficiently close to  $m$ : a contrary sequence would have a subsequence converging to a captured central root. Hence every nearby point of  $E$  lies in this finite raw image union  $Z$ . This family need not enumerate escaping pairs. One bounded surviving pair per target gives the containment.

Discard germs whose central product differs from  $s_0$ . If there are any, let  $d > 0$  be the minimum of their finitely many distances  $|S_i(0) - s_0|$ . Shrink the parameter disk so  $|S_i(t) - S_i(0)| < d/3$  and restrict the target product to  $|s - s_0| < d/3$ . None of those germs can represent a target in that product disk. If there are no discarded germs, the step is vacuous. Every retained branch has center  $y_0$ .

Stabilize these finitely many product comparisons as before, with parameter radius  $\delta_0$ . Shrink the capture neighborhood  $U$  into  $\{|p - m| < \delta_0^N\}$ . Then

$$E \cap U \subset Z_0, \quad Z_0 = \bigcup_j \gamma_j(\{|t| < \delta_0\}),$$

because every represented parameter over  $U$  has  $|t| < \delta_0$ . Choose  $0 < \delta < \delta_0$  so every retained full raw  $\delta$ -image lies in  $U$ , and put

$$W = U \cap \{|p - m| < \delta^N\}, \quad Z_j = \gamma_j(\{|t| < \delta\}).$$

Each  $Z_j$  lies in  $W$ . Conversely, a target in  $W$  represented in an old  $\delta_0$ -image has  $|t|^N < \delta^N$ , hence  $|t| < \delta$ . Using relative closedness in the old value cylinder and the localized containment  $E \cap U \subset Z_0$ , we obtain

$$H \cap W \subset \bigcup_j Z_j, \quad Z_j \subset W. \quad (7.2)$$

Indeed, a sequence from  $E$  converging to a point of  $W$  is eventually in  $U$ , and its represented parameters have a subsequence strictly inside the old disk. The limit's value then places its parameter inside the new disk. Matching the value radius with  $\delta^N$  is essential to retain each whole raw  $\delta$ -image inside  $W$ .

Write  $Z_j^* = \gamma_j(\{0 < |t| < \delta\})$ . These punctured images are connected, pairwise equal or disjoint, and locally smooth graphs over the value. They avoid  $p = m$ . Their intersections with  $H$  are closed relative to  $Z_j^*$ . They are also open, as follows.

At a point  $y \in H \cap Z_j^*$ , (6.1) supplies a complete analytic subbranch through  $y$  contained in  $H$ . Shrink it inside  $W$ . By (7.2) and the local single-graph description of the finite  $Z$  union, it is contained in the graph of  $Z_j^*$  near  $y$ . Its value coordinate is nonconstant. The open mapping theorem makes its value image a neighborhood of the value at  $y$ ; the graph has one product coordinate per value, so the subbranch fills a neighborhood of  $y$  in  $Z_j^*$ . This proves openness of membership in  $H$ .

Connectedness now gives  $Z_j^* \subset H$  or  $Z_j^* \cap H = \emptyset$ . All representatives lie inside  $W$ , so the capture remains valid along the whole connected punctured image. Since  $y_0 \in \text{cl } E$  and  $E$  contains no point of value  $m$ , at least one of the finitely many punctured branches meets  $E$  arbitrarily close to  $y_0$ . Its punctured image is contained in  $H$ , and closedness includes its center. This selection uses membership in  $H$ , not a count attached to the partial bounded-survivor family. Every omitted-value point of  $H$  therefore lies on a complete branch contained in  $H$ ; no isolated center is added.

## 8 Nonconstant product coordinates and the global contradiction

No actual image germ used above has a nonzero constant product. Suppose  $z(t)w(t) = s_0 \neq 0$  on such a germ. The common-value coordinate makes  $z(t)$  nonconstant: a constant physical coordinate would have constant common value, whereas the value parameter is nonconstant. Choose a noncentral parameter with finite common value. Both physical coordinates then avoid zeros of the respective entire factors. The open mapping theorem applied to  $z(t)$  gives

$$A(z) = B(s_0/z)$$

on an open set where both factors are nonzero. There,

$$\frac{zP'_{s_0}(z)}{P_{s_0}(z)} = \frac{zg'(z)}{g(z)} - \frac{(s_0/z)(g^\#)'(s_0/z)}{g^\#(s_0/z)} = A(z) - B(s_0/z) = 0.$$

The identity theorem makes  $P'_{s_0}$  vanish throughout  $\mathbb{C}^\times$ . This contradicts Lemma 3. For a germ centered at poles, its noncentral common values are finite, so the same argument applies. No regularity of the total image at that parameter is required.

The complete branches in  $H$  at ordinary finite, omitted, and infinite values therefore have nonconstant analytic product coordinates. The open mapping theorem makes the product projection of  $H$  open, including at branch centers. The projection is closed because  $H$  is closed in  $\mathbb{C} \times \widehat{\mathbb{C}}$  and  $\widehat{\mathbb{C}}$  is compact. If  $E$  were nonempty, this projection would be a nonempty open-and-closed subset of the connected plane, hence the whole plane. But the cylinder  $|s| < \varepsilon^2$  is disjoint from  $H$ . This contradiction proves the following assertion.

**Proposition 5.** *Every regular finite-valued target in  $Y$  has at most  $L$  distinct physical pairs in its fiber; equivalently,  $E = \emptyset$ .*

## 9 Countable exceptions and arbitrarily large good radii

Let  $D$  consist of the nonregular actual image targets in  $\mathbb{C}^\times \times (\mathbb{C} \setminus \{m\})$ . The finite local family shows that  $D$  is locally finite: all noncentral points in a small local image cover are single-graph points. At an empty central fiber, properness supplies a neighborhood outside the image. The ambient space is second countable; a countable cover by neighborhoods meeting  $D$  finitely shows that  $D$  is countable. Its product projection is countable, although that projection need not be locally finite.

Let  $Q$  be the positive radii  $r$  for which  $(r^2, p) \in D$  for some finite  $p$ . This is countable since  $D$  is countable and  $r \mapsto r^2$  is injective on positive radii. For  $r > 0$  outside  $Q$  and any real  $p > m$ , a nonempty fiber  $F(r^2, p)$  lies at a regular finite target. Properness proves it finite, and Proposition 5 bounds its cardinality by  $L$ . An empty fiber is finite with cardinality zero. No convention for the cardinality of an infinite set is used.

Set  $G(r) = \max_{|z|=r} |g(z)|^2$  for  $r > 0$ . The maximum principle makes  $G$  strictly increasing: equality at two radii would put a maximum point from the smaller circle in the interior of the larger disk, forcing  $g$  to be constant. Also  $G(r) \geq |g(0)|^2 > 0$ .

Local Lipschitz continuity has a direct uniform estimate. Fix  $0 < a < b$ . On  $[a, b]$ , the gradient norm of  $|g|^2$  is  $2|g||g'|$ , bounded by a constant  $C$  on  $|z| \leq b$ . If  $z$  is a maximum point at radius  $r \in [a, b]$ , compare it with  $(u/r)z$  at radius  $u \in [a, b]$ . Their distance is  $|r - u|$ , and the segment between them lies in that disk. The gradient bound gives  $G(r) \leq G(u) + C|r - u|$ . Reversing the radii gives  $|G(r) - G(u)| \leq C|r - u|$ .

Thus  $G$  is absolutely continuous on every compact positive interval, and

$$\int_a^b G'(u) du = G(b) - G(a) > 0.$$

Monotonicity gives  $G' \geq 0$  almost everywhere. The set of radii inside  $(a, b)$  where  $G$  is differentiable and  $G' > 0$  therefore has positive measure. Removing the countable set  $Q$  leaves such a radius  $r \notin Q$ .

At every maximum point  $z$  of  $g$  on this circle,  $g(z) \neq 0$ . Differentiating  $\theta \mapsto |g(e^{i\theta}z)|^2$  at its maximum gives angular stationarity. The smooth supporting function  $h_z(u) = |g(uz/r)|^2$  satisfies  $h_z(u) \leq G(u)$  and  $h_z(r) = G(r)$ . Since both functions are differentiable at  $r$ , their derivatives agree there. This applies to every maximizing  $z$ , even when the maximum is tied. Consequently

$$\operatorname{Im} \frac{zg'(z)}{g(z)} = 0, \quad \operatorname{Re} \frac{zg'(z)}{g(z)} = \frac{rG'(r)}{2G(r)} > 0.$$

All maximum points therefore have the same real actual value

$$A(z) = p = m + \frac{rG'(r)}{2G(r)} > m.$$

The injective map  $z \mapsto (z, \bar{z})$  sends them into the explicitly finite fiber  $F(r^2, p)$ , because  $B(\bar{z}) = \overline{A(z)} = p$ . Their cardinality is at most  $L = 2k$ . The maximum-point sets of  $f$  and  $g$  coincide, so the same bound holds for  $f$ . Choosing the interval  $(R, R + 1)$  proves Theorem 1.

## 10 Formalization and verification

The accompanying Lean 4 project proves the theorem with the original entire, nonzero, and non-monomial hypotheses. Its final declaration is

`MaximumModulus.bounded_maxPoints_at_arbitrarily_large_radii`

in `MaximumModulus/AllRadii.lean`, line 26. The statement spells out the set using the norm equality and all actual function-value comparisons. It has one  $B : \mathbb{N}$  before the universal quantifier over  $R : \mathbb{R}$ , and explicitly supplies finiteness together with the bound on `Set.ncard`. Thus the convention for `Set.ncard` on an infinite set cannot make the result vacuous. Independent finiteness on every positive circle is in `MaximumModulus/Finiteness.lean`, line 187. The negative-growth implication is in `MaximumModulus/AllRadii.lean`, line 36.

The principal correspondence between the proof and project modules is as follows. All filenames in the table are relative to `MaximumModulus/`.

| Mathematical step                            | Lean source modules                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Circle finiteness and logarithmic derivative | <code>Finiteness.lean</code> , <code>LogDerivative.lean</code>                                                 |
| Local power coordinates and small fibers     | <code>LocalValence.lean</code> , <code>SmallFibers.lean</code>                                                 |
| Exact finite covers and distinct-pair counts | <code>FiniteTargetCover.lean</code> , <code>DuplicateSafeCount.lean</code>                                     |
| Ordinary finite values and infinity          | <code>HighCountInterior.lean</code> , <code>InfinityInterior.lean</code>                                       |
| Omitted-value capture and continuation       | <code>BoundaryCharts.lean</code> , <code>BranchContinuation.lean</code> , <code>OmittedTargetCover.lean</code> |
| Global empty high-count locus                | <code>HighCountGlobal.lean</code>                                                                              |
| Countable exceptions and radial envelope     | <code>FiniteExceptionalSet.lean</code> , <code>MaximumEnvelope.lean</code>                                     |
| Final statement and negative implication     | <code>AllRadii.lean</code>                                                                                     |

The crucial exact-cover, central-count, and omitted-value assertions are proved by the declarations

```
sphereCorrespondence_regular_finite_family,
central_fiber_count_le_of_total_graph,
omittedHighCountBranchCover_exists,
OmittedHighCountBranchCover.actual_closure_branches,
regular_high_count_closure_omitted_local_projection_isOpen,
regularLogDerivativeHighCount_empty.
```

The exact toolchain and dependency revisions are

```
Lean toolchain: leanprover/lean4:v4.35.0-rc4
Lean commit: c29b6dda4f7c20e3eeaa717c4e565663c5cfa364
mathlib commit: 81d17696471311f5e3e2f034236df7a781449f7e.
```

All transitive dependency revisions are pinned in `lake-manifest.json`. Build and verification instructions are in the project `README.md`; `sh scripts/verify.sh` performs the recorded project checks.

A clean project build, direct checks of all project sources, sequential kernel replays of the 42 project modules relative to their pinned imports, and explicit target axiom inspections have been executed successfully. The final theorem and its transitive dependencies use only `propext`, `Classical.choice`, and `Quot.sound`. They contain no proof placeholders or unproved custom mathematical axioms. The requested prize verification workflow was also executed locally on a frozen audit snapshot. Eight theorem and independent bridge targets passed; the independently specified final statement and exported solution closure were accepted by the configured comparison and external checker runs. These checks concern the formal theorem; they do not constitute a publication or prize decision.

Full commands, logs, axiom reports, pins, comparison outputs, and isolation details are preserved in `verification/prize-report.md` and its linked evidence. The audit snapshot commit `8b731b51dcb0ed3d7e8c6463b6266c35d8099c85` is a temporary local verification record,

not a published repository revision. The mathematical exposition here was revised after those checks without changing the verified Lean source, dependency lock, or build scripts.

## 11 Scope and differences from the earlier manuscript

The result proved here is the cofinal finite bound in Theorem 1, which suffices for the original all-radii question. It remains compatible with unbounded counts along selected radii. The earlier manuscript also proposed a locally finite exceptional set of radii for a  $2k$  bound and sharpness examples. Those stronger assertions are outside the verified scope of this paper and its final Lean theorem. In particular, local finiteness of the nonregular *target* set  $D$  does not by itself imply local finiteness of its projection onto radii. The argument in Section 9 uses countability and a positive-measure set of radii; it is not an assertion of an almost-everywhere maximum-point bound.

The continuation proof replaces the earlier proposed propagation of global irreducible-component degrees by the closure of actual regular high-count targets. The replacement makes explicit the total-image graph condition, the exact finite cover, the central distinct-pair inequality, simultaneous target and parameter shrinking, and the reciprocal coordinate at infinity. At the omitted value it captures a bounded surviving pair for each target and derives branch selection from closed and open membership in  $H$ . It never assumes that the partial family counts escaping pairs. The original manuscript has been preserved separately without alteration.

**Assistance disclosure.** OpenAI’s ChatGPT and Codex (GPT-6) were used to develop the proof and prepare the manuscript. The human author takes full responsibility for the mathematical arguments and the final manuscript. Codex also assisted with the Lean formalization. The accompanying verification records describe the checks actually performed and their scope.

## References

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<https://arxiv.org/pdf/1809.07200v2#page=30>.
- [2] F. Herzog and G. Piranian, *The counting function for points of maximum modulus*, in *Entire Functions and Related Parts of Analysis* (Proc. Sympos. Pure Math., La Jolla, 1966), American Mathematical Society, Providence, RI, 1968, pp. 240–243.
- [3] Adi Glücksam and Leticia Pardo-Simón, *An approximate solution to Erdős’ maximum modulus points problem*, arXiv:2208.11154v2 (2023), Question 1.1.  
<https://arxiv.org/html/2208.11154v2#S1>.
- [4] Leticia Pardo-Simón and David J. Sixsmith, *A finite-order answer to a problem of Erdős on maximum modulus points*, arXiv:2607.09462v1 (2026), introduction.  
<https://arxiv.org/html/2607.09462v1#S1>.